

An Example Related to Gregory's Theorem

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Abstract

In this paper, we give an example of a complete, computable infinitary theory T with countable models $\mathcal{M}$ and $\mathcal{N}$, where $\mathcal{N}$ is a proper computable infinitary extension of $\mathcal{M}$ and T has no uncountable model. In fact, $\mathcal{M}$ and $\mathcal{N}$ are (up to isomorphism) the only models of T . Moreover, for all computable ordinals α , the computable Σ_α part of T is hyperarithmetical. It follows from a theorem of Gregory [8], [7] that if T is a Π_1^1 set of computable infinitary sentences and T has a pair of models $\mathcal{M}$ and $\mathcal{N}$, where $\mathcal{N}$ is a proper computable infinitary extension of $\mathcal{M}$, then T would have an uncountable model.

1 Introduction

In the 1960's and the 1970's, there were investigations of the fine structure of the constructible sets, of the model theory of infinitary logic, and of generalizations of computability theory (" α -recursion"). This work combined ideas from set theory, model theory, and computability. Jon Barwise [4] proved a version of Compactness for infinitary logic, and he became a great spokesperson for the unity of logic. There were many contributors to the body of work, including Georg Kreisel [14], Saul Kripke, Richard Platek, Carole Karp [10], Ronald Jensen [9], John Schlipf, Jean-Pierre Ressayre [16], and Gerald Sacks [18] and several of his students, including Sy Friedman [5], Richard Shore, [19], and Ted Slaman [20]. There is a body of recent work, again connecting different branches of logic. In connection with some of this work [2], [3], John Baldwin asked to see an example of the kind we construct.

In the remainder of the introduction, we give some background on hyperarithmetical sets and on infinitary logic, and we state the Barwise Compactness Theorem, and the theorem of Gregory. We describe the computable infinitary formulas. In Section 2, we construct our example of a complete computable infinitary theory T having (up to isomorphism) just two models $\mathcal{M}$ and $\mathcal{N}$, where $\mathcal{N}$ is a proper computable infinitary extension of $\mathcal{M}$. For each computable ordinal α , the Σ_α part of T is hyperarithmetical, since, as we shall see, it is satisfied in a hyperarithmetical substructure of $\mathcal{M}$.

*The authors are grateful to John Baldwin, Sy Friedman, and Gerald Sacks for their helpful and encouraging comments.

1.1 Background on hyperarithmetical sets

We give just a little background here on the hyperarithmetical sets. For more information, see [17], [1], [12]. We write φ_e for the partial function computed by program number e on an effective list. Similarly, we write φ_e^X for the partial function computed using program number e with oracle X . We write $\varphi_e(n) \downarrow$ if program e eventually halts, given input n . Similarly, we write $\varphi_e^X(n) \downarrow$ if interactive program number e eventually halts, given oracle X and input n .

- The *halting set* is $K = \{e : \varphi_e(e) \downarrow\}$. This set is computably enumerable (or c.e.) but not computable.
- For an arbitrary set X , the *jump* is $X' = \{e : \varphi_e^X(e) \downarrow\}$. This set is c.e. relative to X but not computable relative to X .

We can iterate the jump to get $X^{(n)}$, for all $n \in \omega$.

- $X^{(0)} = X$
- $X^{(n+1)} = (X^{(n)})'$

We can continue the iteration process through the computable ordinals.

- Let $X^{(\omega)} = \{ \langle n, x \rangle : x \in X^{(n)} \}$.
- $X^{(\alpha+1)} = (X^{(\alpha)})'$
- for limit α , $X^{(\alpha)}$ represents $\{ \langle \beta, x \rangle : x \in X^{(\beta)} \}$. To make this precise, we must code the ordinals by natural numbers, using Kleene's system of ordinal notation. We shall ignore this point.

Stephen Kleene and Andrzej Mostowski independently defined what is now called the *arithmetical hierarchy*. Martin Davis and Andrzej Mostowski independently defined what is now called the *hyperarithmetical hierarchy*, extending the arithmetical hierarchy through the “computable” ordinals. We define the two hierarchies in a way that is uniform.

- **Arithmetical Hierarchy.** For $1 \leq n < \omega$, a set is Σ_n^0 if it is computably enumerable relative to $\emptyset^{(n-1)}$. A set is Π_n^0 if the complement is Σ_n^0 . A set is Δ_n^0 if it is both Σ_n^0 and Π_n^0 .
- **Hyperarithmetical hierarchy, extending the arithmetical hierarchy.** For a computable ordinal $\alpha \geq \omega$, a set is Σ_α^0 if it is c.e. relative to $\emptyset^{(\alpha)}$. A set is Π_α^0 if the complement is Σ_α^0 . A set is Δ_α^0 if it is both Σ_α^0 and Π_α^0 .

There are only countably many computable ordinals, and there are only countably many hyperarithmetical sets.

1.2 Background on infinitary logic

The infinitary logic $L_{\omega_1\omega}$ allows countable disjunctions and conjunctions, but only finite tuples of quantifiers. We say only a little about $L_{\omega_1\omega}$ here. For more information, see the beautiful book by Keisler [13]. While Keisler allows formulas with infinitely many free variables, our formulas will have only finitely many free variables (as in [1]). There is no prenex normal form for formulas of $L_{\omega_1\omega}$. In general, we cannot bring quantifiers to the front. However, we can bring negations inside, and this results in a kind of normal form. We restrict our attention to formulas in this normal form. We classify these formulas as Σ_α or Π_α for countable ordinals α .

Classification of formulas of $L_{\omega_1\omega}$

1. $\varphi(\bar{x})$ is Σ_0 and Π_0 if it is finitary quantifier-free,
2. for a countable ordinal $\alpha > 0$,
 - $\varphi(\bar{x})$ is Σ_α if it is a countable disjunction of formulas $(\exists \bar{u}) \psi(\bar{x}, \bar{u})$, where ψ is Π_β for some $\beta < \alpha$,
 - $\varphi(\bar{x})$ is Π_α if it is a countable conjunction of formulas $(\forall \bar{u}) \psi(\bar{x}, \bar{u})$, where ψ is Σ_β for some $\beta < \alpha$.

Negations. For each formula φ (in normal form), there is a formula in normal form that is logically equivalent to the negation of φ . This formula, denoted by $\text{neg}(\varphi)$, is obtained by switching disjunction/exists with conjunction/for all, and switching atomic formulas with their negations.

Computable infinitary formulas

The *computable* infinitary formulas are formulas of $L_{\omega_1\omega}$ in which the infinite disjunctions and conjunctions are over c.e. sets. To make this precise, we would assign indices to the formulas based on Kleene's system of ordinal notation, as is done in [1]. Since the infinite disjunctions and conjunctions in a computable infinitary formula are c.e., the formula is in some way "comprehensible." We classify computable infinitary formulas as *computable* Σ_α or *computable* Π_α , for computable ordinals α .

Recall that for finite $n \geq 1$, a set is Σ_n^0 in the arithmetical hierarchy just in case it is definable in the standard model of arithmetic by a finitary Σ_n formula. This fact extends as follows.

Proposition 1.1. *For computable ordinals $\alpha \geq 1$, a set is Σ_α^0 in the hyperarithmetical hierarchy just in case it is definable in the standard model of arithmetic by a computable Σ_α formula.*

The computable infinitary formulas are also connected with the effective Borel hierarchy [21].

Theorem 1.2 (Vanden Boom). *Let K be a class of structures, all with universe ω , and all having a fixed computable language. Suppose K is closed under isomorphism. Then K is Σ_α in the effective Borel hierarchy iff it is axiomatized by a computable Σ_α sentence.*

Admissible sets and admissible fragments of $L_{\omega_1\omega}$

An *admissible set* is a transitive model of *Kripke-Platek set theory*, where this is a weakening of ZF in which the power set axiom is dropped and the separation and collection axioms are restricted to formulas with only bounded quantifiers. Some people also drop the Axiom of Infinity, but we shall keep it.

Definition 1.1. *Let A be a countable admissible set.*

- *The admissible fragment L_A consists of the $L_{\omega_1\omega}$ formulas that are elements of A .*
- *A set is Σ_1 on A if it is defined in (A, ϵ) by a (finitary) formula with existential quantifiers in front, and then only bounded quantifiers.*
- *A set X is A -finite if $X \in A$.*

The least non-computable ordinal is *Church-Kleene* ω_1 , denoted by ω_1^{CK} . The least admissible set is $L_{\omega_1^{CK}}$. For $A = L_{\omega_1\omega}$, we have the following.

- The L_A formulas are essentially the computable infinitary formulas.
- A set or relation on ω is Σ_1 on A iff it is Π_1^1 .
- A set or relation on ω is A -finite iff it is hyperarithmetical.

We come to the Barwise Compactness Theorem.

Theorem 1.3 (Barwise Compactness). *Let A be a countable admissible set, and let T be a set of L_A sentences that is Σ_1 on A . If every A -finite subset of T has a model, then T has a model.*

Here is the special case of Barwise Compactness for the case where A is the least admissible set. Kreisel, in an obscure footnote [14], gave a result of this kind for “ ω -logic”.

Theorem 1.4 (Barwise-Kreisel Compactness). *Suppose T is a Π_1^1 set of computable infinitary sentences. If every hyperarithmetical subset of T has a model, then T has a model.*

Ressayre [15], [16] defined a version of saturation that goes with Barwise Compactness.

Definition 1.2 (Σ -saturation). *Let A be a countable admissible set. A structure $\mathcal{M}$ is Σ_A -saturated if it satisfies the conditions below.*

1. Let $\Gamma(\bar{a}, x)$ be a set of L_A formulas that is Σ_1 on A , where $\bar{a}$ is a tuple in $\mathcal{M}$. If every A -finite set $\Gamma'(\bar{a}, x) \subseteq \Gamma(\bar{a}, x)$ is satisfied in $\mathcal{M}$, then there is some element satisfying the whole set.
2. Let I be an A -finite set, let $\bar{a}$ be a tuple in $\mathcal{M}$, and let Λ be a set of pairs (i, φ) , where $i \in I$ and $\varphi(\bar{a})$ is an L_A formula. We write Λ_i for $\{\varphi(\bar{a}) : (i, \varphi(\bar{a})) \in \Lambda\}$, and similarly, for $\Lambda' \subseteq \Lambda$, we write Λ'_i . Suppose that Λ is Σ_1 on A , and for every A -finite set $\Lambda' \subseteq \Lambda$, there is some $i \in I$ such that $\mathcal{M}$ satisfies Λ'_i . Then there is some $i \in I$ such that $\mathcal{M}$ satisfies Λ_i .

In the definition of Σ -saturation, the second condition is used in the result below, for witnessing disjunction.

Theorem 1.5 (Ressayre). *Suppose $\mathcal{M}$ is a countable Σ_A -saturated L -structure. Suppose T is a set of L'_A sentences that is Σ_1 on A , where L' is a finite extension of L . If the consequences of T in the language L are all true in $\mathcal{M}$, then $\mathcal{M}$ can be expanded to a Σ_A -saturated model of T .*

Remark. In Ressayre's Theorem, we allow the case where $\mathcal{M}$ sits as a relativized reduct of the expansion $\mathcal{N}$.

Notation.

- For an arbitrary admissible set A , we write $\mathcal{M} \preceq_{L_A} \mathcal{N}$ if $\mathcal{M}$ is an L_A -elementary substructure of $\mathcal{N}$; i.e., for all L_A formulas $\varphi(\bar{u})$ and all $\bar{a}$ in $\mathcal{M}$, $\mathcal{M} \models \varphi(\bar{a})$ iff $\mathcal{N} \models \varphi(\bar{a})$. Similarly, we write $\mathcal{M} \prec_{L_A} \mathcal{N}$ if $\mathcal{M}$ is a proper L_A -elementary substructure of $\mathcal{N}$.
- For the special case where A is the least admissible set, we write $\mathcal{M} \preceq_\infty \mathcal{N}$, or $\mathcal{M} \prec_\infty \mathcal{N}$.
- For a computable ordinal α , we write $\mathcal{M} \preceq_\alpha \mathcal{N}$, or $\mathcal{M} \prec_\alpha \mathcal{N}$ if satisfaction of computable Σ_α formulas is the same in the two structures.

For later use, we mention two easy results, both proved in the same way as their finitary analogues (see [13]).

Proposition 1.6 (Tarski Criterion). *Let A be an admissible set. Suppose $\mathcal{M}$ is a substructure of $\mathcal{N}$ and for all L_A formulas φ , if $\mathcal{N} \models \varphi(\bar{c}, b)$, where $\bar{c} \in \mathcal{M}$, then there exists $b' \in \mathcal{M}$ such that $\mathcal{N} \models \varphi(\bar{c}, b')$. Then $\mathcal{M} \preceq_{L_A} \mathcal{N}$.*

Remark. Proposition 1.6 holds for the least admissible set, where the L_A formulas are the computable infinitary formulas and $\preceq_{L_A}$ is $\preceq_\infty$. It also holds if we restrict to computable Σ_α formulas and replace $\preceq_\infty$ by $\preceq_\alpha$.

Proposition 1.7 (Unions of chains). *Let A be an admissible set. Suppose $(\mathcal{M}_n)_{n \in \omega}$ is a chain of L -structures, with union $\mathcal{M}$. Suppose that for all n , $\mathcal{M}_n \preceq_{L_A} \mathcal{M}_{n+1}$. Then for all n , $\mathcal{M}_n \prec_{L_A} \mathcal{M}$.*

Remark. Proposition 1.7 holds for $\preceq_\infty$, and for $\preceq_\alpha$.

1.3 Gregory's Theorem

Here is Gregory's Theorem [8], [7].

Theorem 1.8 (Gregory). *Let A be a countable admissible set. Suppose T is a set of L_A sentences that is Σ_1 on A . If T has a pair of countable models $\mathcal{M}$ and $\mathcal{N}$ such that $\mathcal{M} \prec_{L_A} \mathcal{N}$ ($\mathcal{N}$ is a proper L_A -elementary extension of $\mathcal{M}$), then T has an uncountable model.*

The statement of Gregory's Theorem is reminiscent of the two-cardinal theorem of Vaught that figured in Morley's Categoricity Theorem. However, in Vaught's proof, the uncountable model (in which the interpretation of a particular formula is countable) is the union of an elementary chain of countable models $(\mathcal{M}_\alpha)_{\alpha < \omega+1}$, where all $\mathcal{M}_\alpha$ are isomorphic since they are homogeneous and realize the same types. The proof of Gregory's Theorem is necessarily different, since the infinitary theory may require that all of the elements realize different types—the example that we give in Section 2 illustrates this. Gregory's original proof was quite clever. In a review [11] of Gregory's paper, written ten years after the paper was published, the second author gave a simpler proof, using the notion of Σ -saturation. This proof is sketched below.

Proof of Gregory's Theorem. Suppose T has a model with a proper L_A -elementary extension. Then there are countable models $\mathcal{M}'$ and $\mathcal{N}'$ such that $\mathcal{M}' \prec_{L_A} \mathcal{N}'$ and the structure $(\mathcal{N}', \mathcal{M}')$, with a predicate for the universe of the smaller structure, is Σ_A -saturated. Since $\mathcal{M}'$ has a proper L_A -elementary extension, we can see that $\mathcal{N}'$ satisfies the consequences of having a proper L_A -elementary extension. For any countable Σ_A -saturated model of T plus the consequences of having a proper L_A -elementary extension, we can apply Ressayre's Theorem to get a proper L_A -elementary extension that is Σ_A -saturated. Thus, we can form an L_A -elementary chain $(\mathcal{M}_\alpha)_{\alpha < \omega_1}$ of models of T . For limit α , we let $\mathcal{M}_\alpha = \cup_{\beta < \alpha} \mathcal{M}_\beta$. The union of an L_A elementary chain of Σ_A -saturated models is again Σ_A -saturated. Then $\mathcal{M} = \cup_{\alpha < \omega_1} \mathcal{M}_\alpha$ is an uncountable model of T . □

Gregory stated that there were known examples showing that the assumption that T is Σ_1 on A cannot be dropped. He did not give an example. We have been unable to find one published, and none of the people we have asked remembers seeing one. In the next section, we construct an example, using iterated forcing.

1.4 α -generic sets and families

Let α be a computable ordinal. Recall that $2^{<\omega}$ denotes the set of functions from finite subsets of ω to 2. For a given set I , a relation $R \subseteq I \times \omega$ represents a family of subsets of ω indexed by the elements of I —the set with index i is $R_i = \{n : (i, n) \in R\}$. We write $2^{<I \times \omega}$ for the set of functions from finite subsets of $I \times \omega$ to 2.

Definition 1.3.

- A set $X \subseteq \omega$ is α -generic if for each Σ_α^0 set $D \subseteq 2^{<\omega}$, there is some $p \subseteq \chi_X$ such that either $p \in D$ or else there is no $q \supseteq p$ such that $q \in D$.
- Let I be a computable set of indices. An indexed family $R \subseteq I \times \omega$ is α -generic if for each Σ_α^0 set $D \subseteq 2^{<I \times \omega}$, there is some $p \subseteq \chi_R$ such that either $p \in D$ or else there is no $q \supseteq p$ with $q \in D$.

Fact. Let α be a computable ordinal. There is an α -generic set X such that X is $\Delta_{\alpha+1}^0$. The same is true for an indexed family, with computable index set.

Definition 1.4. Suppose $X, Y \subseteq \omega$. We say that Y is α -generic over X if for all $D \subseteq 2^{<\omega}$ such that D is Σ_α^0 relative to X , there is some $p \subseteq \chi_Y$ such that either $p \in D$ or else there is no $q \supseteq p$ such that $q \in D$. In a similar way, we define what it means for one indexed family to be α -generic over another.

Lemma 1.9. Suppose $\alpha < \beta$. If Y is β -generic, or β -generic over X , then it is α -generic, or α -generic over X . This is true also for indexed families.

In the next section, we shall combine families with disjoint index sets.

Lemma 1.10. Suppose I and J are disjoint computable sets. Let $R \subseteq I \times \omega$ be α -generic, and suppose $R' \subseteq J \times \omega$ is α -generic over R . Then $R \cup R'$ is α -generic.

Proof. For R , we use $2^{<I \times \omega}$ as the set of forcing conditions; for R' , we use $2^{<J \times \omega}$; and for $R \cup R'$, we use $2^{<(I \cup J) \times \omega}$. Let D be a Σ_α^0 subset of $2^{<(I \cup J) \times \omega}$. Each $p \in D$ is the union of a “first part” $p_1 \in 2^{<I \times \omega}$ and a “second part” $p_2 \in 2^{<J \times \omega}$. Let

$$D^2 = \{p_2 \in 2^{<J \times \omega} : (\exists p_1 \in 2^{<I \times \omega}) (p_1 \cup p_2 \in D \text{ \& } p_1 \subseteq \chi_R)\};$$

i.e., D^2 is the set of second parts of conditions in D with first part extending to χ_R . This is Σ_α^0 relative to R . There is some $p_2^* \subseteq \chi_{R'}$ such that either $p_2^* \in D^2$ or else p_2^* has no extension in D^2 .

Case 1: Suppose $p_2^* \in D^2$. We have $p_1 \subseteq \chi_R$ such that $p_1 \cup p_2^* \in D$, and $p_1 \cup p_2^* \subseteq \chi_{R \cup R'}$.

Case 2: Suppose p_2^* has no extension in D^2 . Let

$$D_1 = \{p_1 \in 2^{<I \times \omega} : p_1 \cup p_2^* \in D\}.$$

This is Σ_α^0 . For $p_1 \subseteq \chi_R$, $p_1 \notin D_1$. Therefore, we have $p_1^* \subseteq \chi_R$, such that p_1^* has no extension in D_1 . Then $p_1^* \cup p_2^* \subseteq \chi_{R \cup R'}$, and this has no extension in D . $\square$

Definition 1.5. For ordinals α and β , we say that β is much greater than α if $\alpha + 1 + \beta = \beta$.

For example, if $\alpha = \omega$, then $\beta = \omega^2$ is much greater than ω .

Remark. Suppose α and β are computable ordinals such that β is much greater than α . In this case, a set, or family of sets, that is Σ_β^0 relative to a $\Delta_{\alpha+1}^0$ set is just Σ_β^0 .

We combine the results above to get the following simple proposition, which we shall use in the next section.

Proposition 1.11. *Suppose α and β are computable ordinals, with β much greater than α . Let I and J be disjoint computable sets. Suppose $R \subseteq I \times \omega$ is α -generic and $\Delta_{\alpha+1}^0$, and suppose $R' \subseteq J \times \omega$ is β -generic. Then $R \cup R'$ is α -generic.*

Proof. By the remark above, R' is β -generic over R . By Lemma 1.9, R' is α -generic over R . By Lemma 1.10, $R \cup R'$ is α -generic. □

2 The example

In this section, we prove the following.

Theorem 2.1. *There is a set T of computable infinitary sentences, in a computable language L , such that T has (up to isomorphism) exactly two models, $\mathcal{M}$ and $\mathcal{N}$, where $\mathcal{M} \prec_\infty \mathcal{N}$. Moreover, for each computable ordinal α , the set of computable Σ_α sentences in T is hyperarithmetical.*

2.1 Outline

We give the general idea of the construction first and then fill in details.

- Let L be the language consisting of unary predicates U_n for $n \in \omega$. This is the language of T .
- For an L -structure $\mathcal{A}$, each element represents a set. The set represented by $x \in \mathcal{A}$ is $\{k \in \omega : \mathcal{A} \models U_k x\}$.
- The universe of $\mathcal{M}$ is an infinite, computable set of constants C , which is partitioned effectively into infinitely many infinite sets C_n . The universe of $\mathcal{N}$ has one extra constant a , in addition to the elements of C .

We will arrange that in models of T , distinct elements represent different sets. The sets represented by the elements of $\mathcal{M}$ are all hyperarithmetical. In all models, the hyperarithmetical sets represented by the elements are the same as for $\mathcal{M}$. We arrange that if there is any other element, it must represent a fixed non-hyperarithmetical set, the one represented by a in $\mathcal{N}$.

We let T be the set of computable infinitary sentences true in $\mathcal{M}$ and $\mathcal{N}$. We choose an increasing sequence of computable ordinals $(\alpha_n)_{n \in \omega}$ with limit ω_1^{CK} . We suppose that α_{n+1} is much greater than α_n .

1. The set represented by the constant a will be *hyperarithmetically generic*; i.e., it is α -generic for all $\alpha < \omega_1^{CK}$.
2. For all $c \in C_n$ and all $k < n$, k will be in the set represented by c iff it is in the set represented by a .
3. Apart from Item 2 above, the sets assigned to the elements of C_n will form an α_n -generic family that is $\Delta_{\alpha_n+1}^0$. Thus, if c is not $\Delta_{\alpha_n+1}^0$, then $c \notin \cup_{m \leq n} C_m$, and the sets represented by c and a must agree on all $k \leq n$.

To prove Theorem 2.1, we must show the following.

Proposition 2.2. $\mathcal{M}$ and $\mathcal{N}$ are (up to isomorphism) the only models of T .

Proposition 2.3. $\mathcal{M} \prec_\infty \mathcal{N}$.

Although we have not yet given the details of the construction, we can prove Proposition 2.2.

Proof of Proposition 2.2. It is enough to note that the theory T consisting of the computable infinitary sentences true in both $\mathcal{M}$ and $\mathcal{N}$ includes the following:

1. a sentence saying that distinct elements differ on some U_k ,
2. sentences saying that all elements representing sets that are not $\Delta_{\alpha_n+1}^0$ satisfy the same predicates U_k for $k \leq n$,
3. sentences saying exactly which hyperarithmetical sets are represented.

We have promised to arrange that the sets represented by a and the elements of C are all distinct. The elements of C_n agree with the set represented by a on $k < n$. The elements of $\cup_{m \leq n} C_m$ will represent sets that are $\Delta_{\alpha_n+1}^0$. Each $c \in \cup_{m > n} C_m$ will represent a set that is at least α_{n+1} -generic, so it cannot be $\Delta_{\alpha_n+1}^0$. In a model of T , the hyperarithmetical sets represented are specified. If there is an element representing a non-hyperarithmetical set then it must agree with a everywhere. Since distinct elements represent different sets, the model must be isomorphic to either $\mathcal{M}$ or $\mathcal{N}$. □

To complete the proof of Theorem 2.1, we must prove Proposition 2.3, saying that $\mathcal{M} \prec_\infty \mathcal{N}$. We need to understand truth in $\mathcal{M}$ and $\mathcal{N}$. We choose a hyperarithmetically generic set A first. We then choose, in succession, families of sets $R_0, R_1, R_2, \dots$ such that

- $R_n \subseteq C_n \times \omega$,
- R_n is α_n -generic,
- R_n is $\Delta_{\alpha_n+1}^0$.

We will let the constant a represent the set A in $\mathcal{N}$. We modify the sets that R_n assigns to elements of C_n to make them agree with A on $k < n$. These modified sets will be the ones represented by the elements of C_n in $\mathcal{M}$ and $\mathcal{N}$. Since A and the families R_n , for $n = 0, 1, 2, \dots$ are chosen separately, our forcing does not let us directly determine truth of more interesting statements in $\mathcal{M}$ or $\mathcal{N}$. We write $C_{\leq n}$ and $C_{< n}$ for the unions of C_k for $k \leq n$, or for $k < n$. Let $\mathcal{M}_n$ be the structure with universe $C_{\leq n}$, and let $\mathcal{N}_n$ be the structure with universe $C_{\leq n} \cup \{a\}$. Our forcing will let us understand truth in these structures so that we can prove the following three propositions.

Proposition 2.4. $\mathcal{M}_n \prec_{\alpha_n} \mathcal{M}_{n+1}$.

Proposition 2.5. $\mathcal{M}_n \prec_{\alpha_n} \mathcal{N}_n$.

Proposition 2.6. $\mathcal{N}_n \prec_{\alpha_n} \mathcal{N}_{n+1}$.

Assuming Propositions 2.4, 2.5, and 2.6, we prove Proposition 2.3 as follows.

Proof. We must show that $\mathcal{M} \prec_{\infty} \mathcal{N}$. Suppose $\mathcal{N} \models \varphi(\bar{c})$, where $\varphi(\bar{x})$ is computable infinitary and $\bar{c} \in \mathcal{M}$. We show that $\mathcal{M} \models \varphi(\bar{c})$. Take n such that $\varphi(\bar{x})$ is computable Σ_{α_n} and $\bar{c} \in \mathcal{M}_n$. Using Proposition 2.4 plus Proposition 1.7, we get $\mathcal{M}_n \prec_{\alpha_n} \mathcal{M}$. Using Proposition 2.6 plus Proposition 1.7, we get $\mathcal{N}_n \prec_{\alpha_n} \mathcal{N}$. By Proposition 2.5, $\mathcal{M}_n \prec_{\alpha_n} \mathcal{N}_n$. Since $\mathcal{M}_n \prec_{\alpha_n} \mathcal{N}_n \prec_{\alpha_n} \mathcal{N}$, $\varphi(\bar{c})$ holds in $\mathcal{N}_n$, and in $\mathcal{M}_n$. Since $\mathcal{M}_n \prec_{\alpha_n} \mathcal{M}$, $\varphi(\bar{c})$ also holds in $\mathcal{M}$. $\square$

We have reduced our main result (Theorem 2.1) to Propositions 2.4, 2.5, and 2.6. To prove the propositions, we must understand precisely how our forcing determines truth in $\mathcal{M}_n$ and $\mathcal{N}_n$.

2.2 A single $\mathcal{M}_n$

The family R_n is chosen using forcing conditions in $2^{<C_n \times \omega}$. We define forcing so as to understand truth of computable Σ_{α_n} statements in $\mathcal{M}_n$. We assume that the sets represented by elements of $C_{< n}$ have been determined, and the set A represented by a is known at least up to n . Suppose $\varphi(\bar{x})$ is a computable Σ_{α_n} formula and $\bar{c} \in C_{\leq n}$. We write $\varphi(\bar{c})$ with the meaning $\mathcal{M}_n \models \varphi(\bar{c})$. We define $p \Vdash \varphi(\bar{c})$, for $p \in 2^{<C_n \times \omega}$, by induction on φ . The most important clauses are the first ones, on forcing of basic statements $\pm U_k c$.

Definition 2.1 (Definition of Forcing).

1. For $c \in C_{< n}$, $p \Vdash U_k c$ if k is in the set represented by c , and $p \Vdash \neg U_k c$ if k is not in this set.
2. For $c \in C_n$, if $k < n$, then $p \Vdash U_k c$ if $k \in A$, and $p \Vdash \neg U_k c$ if $k \notin A$.
3. For $c \in C_n$, if $k \geq n$, then $p \Vdash U_k c$ if $p(c, k) = 1$ and $p \Vdash \neg U_k c$ if $p(c, k) = 0$.

4. Suppose φ is a finitary quantifier-free statement about elements of $C_{\leq n}$. We may suppose that φ is in disjunctive normal form. Then $p \Vdash \varphi$ if there is some disjunct for which p forces all of the conjuncts.
5. Suppose $\varphi(\bar{c}) = \bigvee_i (\exists \bar{u}_i) \varphi_i(\bar{u}_i, \bar{c})$ is computable Σ_β for some $\beta \leq \alpha_n$. Then $p \Vdash \varphi(\bar{c})$ if for some i and some $\bar{d}_i \in C_{\leq n}$, $p \Vdash \varphi_i(\bar{d}_i, \bar{c})$.
6. Suppose $\varphi(\bar{c}) = \bigwedge_i (\forall \bar{u}_i) \varphi_i(\bar{u}_i, \bar{c})$ is computable Π_β for some $\beta \leq \alpha_n$. Then $p \Vdash \varphi$ if for all i , all $\bar{d}_i \in C_{\leq n}$, and all $q \supseteq p$, there is some $r \supseteq q$ such that $r \Vdash \varphi_i(\bar{d}_i, \bar{c})$.

We have the usual lemmas, all proved in a very straightforward way by induction on computable Σ_{α_n} formulas.

Lemma 2.7 (Extension). *If $p \Vdash \varphi$ and $q \supseteq p$, then $q \Vdash \varphi$.*

Lemma 2.8 (Consistency). *It is not the case that $p \Vdash \varphi$ and $p \Vdash \text{neg}(\varphi)$.*

Lemma 2.9 (Density). *Given p and φ , there exists $q \supseteq p$ such that $q \Vdash \varphi$ or $q \Vdash \text{neg}(\varphi)$.*

The next lemma gives an alternative definition of forcing for computable Π_β statements. The proof is very straightforward, using the lemmas above. The lemma lets us see that forcing of computable Σ_β statements is Σ_β^0 and forcing of computable Π_β statements is Π_β^0 .

Lemma 2.10. *For a formula $\varphi(\bar{c})$ of the form $\bigwedge_i (\forall \bar{u}_i) \psi_i(\bar{u}_i, \bar{c})$, we have $p \Vdash \varphi(\bar{c})$ iff for all i , all $\bar{d}_i \in C_{\leq n}$, and all $q \supseteq p$, it is not the case that $q \Vdash \text{neg}(\psi_i(\bar{d}_i, \bar{c}))$.*

Remark. In choosing R_n , we decide all computable Σ_{α_n} statements $\varphi(\bar{c})$. Using Lemma 2.10, we see that the set of conditions forcing $\varphi(\bar{c})$ is $\Sigma_{\alpha_n}^0$. If p has no extension forcing the computable Σ_{α_n} statement $\varphi(\bar{c})$, then p itself forces the computable Π_{α_n} statement $\text{neg}(\varphi(\bar{c}))$.

Lemma 2.11 (Truth-and-Forcing). *For a computable Σ_{α_n} formula $\varphi(\bar{x})$ and a tuple $\bar{c} \in C_{\leq n}$, $\mathcal{M}_n \models \varphi(\bar{c})$ iff there exists $p \subseteq \chi_{R_n}$ such that $p \Vdash \varphi(\bar{c})$.*

The lemmas so far are standard, with standard proofs. We give two further lemmas that are peculiar to our setting. When we defined $p \Vdash \pm U_k c$, for $c \in C_n$ and $k < n$, we ignored $p(c, k)$. Using induction, we easily get the following.

Lemma 2.12 (Reduction to Basic Statements). *Suppose p and q force the same basic statements $\pm U_k c$. Then they force exactly the same computable Σ_{α_n} statements.*

Let f be a permutation of $C_{\leq n}$ that transposes two elements $c, c' \in C_n$, fixing everything else. We write $f(p)$ for the forcing condition that says about the pair c', c (and other elements), what p says about the pair c, c' (and those other elements). For a computable Σ_{α_n} formula $\varphi = \psi(\bar{b}, c, c')$, where the elements of

$\bar{b}, c, c'$ are all distinct, we write $f(\varphi)$ for the formula $\psi(\bar{b}, c', c)$ saying about the tuple $\bar{b}, c', c$ what φ says about the tuple $\bar{b}, c, c'$. Since the basic formulas forced by p about the pair c, c' , are the same as those forced by $f(p)$ about the pair c', c , we get the following, again by induction.

Lemma 2.13 (Transposition). *Suppose f transposes $c, c' \in C_n$, fixing the rest of $C_{\leq n}$. Then for $p \in 2^{<C_n \times \omega}$, for computable Σ_α statements φ , we have $p \Vdash \varphi$ iff $f(p) \Vdash f(\varphi)$.*

2.3 Proof of Proposition 2.4

We must show that $\mathcal{M}_n \prec_{\alpha_n} \mathcal{M}_{n+1}$. For this, it is useful to imagine that R_n and R_{n+1} are chosen together, instead of one after the other. Recall that R_n is an α_n -generic family determined using forcing conditions in $2^{<C_n \times \omega}$, and R_{n+1} is an α_{n+1} -generic family determined using forcing conditions in $2^{<C_{n+1} \times \omega}$. The lemma below says that the union of the two families is α_n -generic, determined using forcing conditions in the set $2^{<(C_n \cup C_{n+1}) \times \omega}$.

Lemma 2.14. *The family $R_n \cup R_{n+1}$ is α_n -generic, where $2^{<(C_n \cup C_{n+1}) \times \omega}$ is the set of forcing conditions.*

Proof. We apply Proposition 1.11. Recall that α_{n+1} is much greater than α_n . The family R_n is α_n -generic and $\Delta_{\alpha_{n+1}}^0$, and the family R_{n+1} is α_{n+1} -generic. Therefore, the family $R_n \cup R_{n+1}$ is α_n -generic. □

We say what it means for forcing conditions in $2^{<(C_n \cup C_{n+1}) \times \omega}$ to force statements about $\mathcal{M}_{n+1}$. We suppose that we have previously determined the sets represented by elements of $C_{<n}$, and we have also determined the set A , at least for $k < n+1$. We consider computable Σ_{α_n} statements $\varphi(\bar{c})$, with the meaning $\mathcal{M}_{n+1} \models \varphi(\bar{c})$, where $\bar{c} \in C_{\leq n+1}$. The important clauses in the definition are the ones for the basic formulas.

Definition 2.2 (Definition of Forcing).

1. For $c \in C_{<n}$, $p \Vdash U_k c$ if k is in the set previously assigned to c , and $p \Vdash \neg U_k c$ if k is not in this set.
2. For $c \in C_n$ and $k < n$, or $c \in C_{n+1}$ and $k < n+1$, $p \Vdash U_k c$ if k is in the set previously assigned to a , and $p \Vdash \neg U_k c$ if k is not in this set.
3. For $c \in C_n$ and $k \geq n$, or $C \in C_{n+1}$ and $k \geq n+1$, $p \Vdash U_k c$ if $p(c, k) = 1$ and $p \Vdash \neg U_k c$ if $p(c, k) = 0$.
4. Suppose φ a finitary quantifier-free statement about elements of $C_{\leq n}$. We suppose that φ is in disjunctive normal form. Then $p \Vdash \varphi$ if there is some disjunct for which p forces all of the conjuncts.
5. Suppose $\varphi(\bar{c}) = \bigvee_i (\exists \bar{u}_i) \varphi_i(\bar{u}_i, \bar{c})$ is computable Σ_β for some $\beta \leq \alpha_n$. Then $p \Vdash \varphi(\bar{c})$ if for some i and some $\bar{d}_i \in C_{\leq n}$, $p \Vdash \varphi_i(\bar{d}_i, \bar{c})$.

6. Suppose $\varphi(\bar{c}) = \bigwedge_i (\forall \bar{u}_i) \varphi_i(\bar{u}_i, \bar{c})$ is computable Π_β for some $\beta \leq \alpha_n$. Then $p \Vdash \varphi$ if for all i , all $\bar{d}_i \in C_{\leq n}$, and all $q \supseteq p$, there is some $r \supseteq q$ such that $q \Vdash \varphi_i(\bar{d}_i, \bar{c})$.

We have the usual lemmas, including Truth-and-Forcing. We also have Reduction to Basic Statements, and Transposition.

Lemma 2.15 (Truth and Forcing). *For a computable Σ_{α_n} formula $\varphi(\bar{x})$ and $\bar{c} \in C_{\leq n+1}$, $\mathcal{M}_{n+1} \models \varphi(\bar{c})$ iff there exists $p \subseteq \chi_{R_n \cup R_{n+1}}$ such that $p \Vdash \varphi(\bar{c})$.*

Lemma 2.16 (Reduction to Basic Statements). *Suppose p, q force the same basic statements. Then they force exactly the same Σ_{α_n} statements.*

Lemma 2.17 (Transposition). *Suppose f transposes $c \in C_{n+1}$ with $c' \in C_n$, fixing the rest of $C_{\leq n+1}$. Suppose p is a forcing condition such that the basic statements forced by p about the pair c, c' (and other elements), are forced by $f(p)$ about the pair c', c (and those other elements). Then for all computable Σ_α statements φ , $p \Vdash \varphi$ iff $f(p) \Vdash f(\varphi)$.*

Suppose, for example, that p mentions c and $\bar{b}$, but not c' , and $p(c, n)$ agrees with a . Then f and p satisfy the conditions of the Transposition Lemma.

We must prove Proposition 2.4, saying that $\mathcal{M}_n \prec \mathcal{M}_{n+1}$. By the Tarski Criterion (Proposition 1.6), it is enough to prove the following.

Proposition 2.4, re-phrased. Suppose $\mathcal{M}_{n+1} \models \varphi(\bar{d}, c)$, where φ is computable Σ_{α_n} , $\bar{d} \in C_{\leq n}$, and $c \in C_{n+1}$. Then there is some $c' \in C_n$ such that $\mathcal{M}_{n+1} \models \varphi(\bar{d}, c')$.

Proof. Suppose $\mathcal{M}_{n+1} \models \varphi(\bar{d}, c)$, where $\varphi(\bar{u}, x)$ is computable Σ_{α_n} , $\bar{d} \in C_n$, and $c \in C_{n+1}$. By the Truth and Forcing Lemma, there is some $p \subseteq \chi_{R_n \cup R_{n+1}}$ such that $p \Vdash \varphi(\bar{d}, c)$. Let p' be the same as p except that for $k \leq n$, $p'(c, k)$ agrees with the set assigned to a . By Lemma 2.16 (Reduction to Basic Statements), $p' \Vdash \varphi(\bar{d}, c)$.

Let f be a permutation of $C_{\leq n+1}$ that transposes c with some $c' \in C_n$, fixing everything else. We choose c' such that $c' \notin \bar{d}$, c' is not mentioned in p or p' , and $\chi_{R_n}(c', k) = p'(c, k)$ for all k such that $p'(c, k)$ is defined. By Lemma 2.17 (Transposition), we have $f(p') \Vdash \varphi(\bar{d}, c')$. Moreover, $f(p') \subseteq \chi_{R_n \cup R_{n+1}}$, so by Lemma 2.15 (Truth and Forcing), $\mathcal{M}_{n+1} \models \varphi(\bar{d}, c')$, as required. $\square$

2.4 Proof of Proposition 2.5

The proof of Proposition 2.5 is similar to that for Proposition 2.4. We must show that $\mathcal{M}_n \prec_{\alpha_n} \mathcal{N}_n$. Here it is useful to imagine that R_n and A are chosen at once. Recall that R_n is obtained using the set of forcing conditions $2^{<C_n \times \omega}$, while $R_A = \{(a, k) : k \in A\}$ is obtained using the set of forcing conditions $2^{\{a\} \times \omega}$. Now, we consider the union as a single α_n -generic family, indexed by $C_n \cup \{a\}$.

Lemma 2.18. $R_n \cup R_A$ is α_n -generic, where $2^{(C_n \cup \{a\}) \times \omega}$ is the set of forcing conditions.

Proof. We apply Proposition 1.11. Recall that α_{n+1} is much greater than α_n . Now, R_n is α_n -generic. Now, A and R_A are α_{n+1} -generic. Therefore, the family $R_n \cup R_A$ is α_n -generic. $\square$

We define forcing. We suppose that sets have been assigned to elements of $C_{<n}$ and A has been determined for $k < n$. We consider computable Σ_{α_n} statements $\varphi(\bar{c}, a)$, for $\bar{c} \in C_{\leq n}$, where the intended meaning is $\mathcal{N}_n \models \varphi(\bar{c}, a)$, for $\bar{c} \in C_{\leq n}$. We define forcing of these statements. The important clauses are the ones for basic statements.

Definition 2.3 (Definition of forcing).

1. For $k < n$, $p \Vdash U_k a$ if $k \in A$, and $p \Vdash \neg U_k a$ if $k \notin A$.
2. For $k \geq n$, $p \Vdash U_k a$ if $p(a, k) = 1$, and $p \Vdash \neg U_k a$ if $p(a, k) = 0$.
3. For $c \in C_{<n}$, $p \Vdash \pm U_k c$ if k is in, or not in, the set assigned to c .
4. For $c \in C_n$ and $k < n$, $p \Vdash U_k c$ if $k \in A$, and $p \Vdash \neg U_k c$ if $k \notin A$.
5. For $c \in C_n$ and $k \geq n$, $p \Vdash U_k c$ if $p(c, k) = 1$, and $p \Vdash \neg U_k c$ if $p(c, k) = 0$.
6. For more complicated formulas, forcing is defined as before.

We have the usual lemmas, including Truth-and-Forcing. We also have Reduction to Basic Statements, and Transposition.

Lemma 2.19 (Truth and Forcing). For a computable Σ_{α_n} formula $\varphi(\bar{u}, x)$ and $\bar{c} \in C_{\leq n}$, $\mathcal{N}_n \models \varphi(\bar{c}, a)$ iff there exists $p \subseteq \chi_{R_n \cup R_A}$ such that $p \Vdash \varphi(\bar{c}, a)$.

Lemma 2.20 (Reduction to Basic Statements). If p, q force the same basic statements, then they force exactly the same computable Σ_{α_n} statements.

Lemma 2.21 (Transposition). Suppose f transposes a with $a' \in C_n$, fixing the rest of $C_{\leq n} \cup \{a\}$. Suppose p is a forcing condition such that the basic statements forced by p about a, a' (and other constants), are forced by $f(p)$ about a', a (and those other constants). Then for all computable Σ_{α_n} statements φ , $p \Vdash \varphi$ iff $f(p) \Vdash f(\varphi)$.

Suppose, for example, that f transposes a with $a' \in C_n$, and p mentions a and some other elements $\bar{b}$, but not a' . Then f and p satisfy the conditions of the Transposition Lemma.

By the Tarski Criterion, to prove that $\mathcal{M}_n \preceq_{\alpha_n} \mathcal{N}_n$, it is enough to show the following.

Proposition 2.5, re-phrased. Suppose $\mathcal{N}_n \models \varphi(\bar{c}, a)$, where $\varphi(\bar{u}, x)$ is computable Σ_{α_n} and $\bar{c} \in C_{\leq n}$. Then there is some $a' \in C_n$ such that $\mathcal{N}_n \models \varphi(\bar{c}, a')$.

Proof. By the Truth and Forcing Lemma, there is some $p \subseteq \chi_{R_n \cup R_A}$ such that $p \Vdash \varphi(\bar{c}, a)$. Let p' be the same as p except that for $k < n$, $p'(a, k)$ agrees with A . By the lemma on Reduction to Basic Statements, $p' \Vdash \varphi(\bar{c}, a)$. Let f be a permutation of $C_{\leq n} \cup \{a\}$ that transposes a with some $a' \in C_n$, fixing everything else. We choose a' such that $a' \notin \bar{c}$, a' is not mentioned in p or p' , and $\chi_{R_n}(a', k) = p'(a, k)$ for all k such that $p'(a, k)$ is defined. By the Transposition Lemma, we have $f(p') \Vdash \varphi(\bar{c}, a')$. Moreover, $f(p') \subseteq \chi_{R_n \cup R_A}$. Then by the Truth and Forcing Lemma, $\mathcal{N}_n \models \varphi(\bar{c}, a')$, as required. $\square$

2.5 Proof of Proposition 2.6

The proof of Proposition 2.6 is similar to the proofs for Propositions 2.4 and 2.5. We must show that $\mathcal{N}_n \prec_{\alpha_n} \mathcal{N}_{n+1}$. For this, it is helpful to imagine that R_n , R_{n+1} , and $R_A = \{(a, k) : k \in A\}$, are chosen all at once, instead of one at a time. Recall that R_n is an α_n -generic family obtained using the set of forcing conditions $2^{<C_n \times \omega}$, R_{n+1} is an α_{n+1} -generic family obtained using the set of forcing conditions $2^{<C_{n+1} \times \omega}$, and R_A is a hyperarithmetically generic set obtained using the set of forcing conditions $2^{<\{a\} \times \omega}$. Now, we consider a single α_n -generic family, indexed by the set $C_n \cup C_{n+1} \cup \{a\}$, and obtained using the set of forcing conditions $2^{(C_n \cup C_{n+1} \cup \{a\}) \times \omega}$. Note that each forcing condition is the union of one element of $2^{<C_n \times \omega}$, one element of $2^{<C_{n+1} \times \omega}$, and one element of $2^{<\{a\} \times \omega}$.

Lemma 2.22. $R_n \cup R_{n+1} \cup R_A$ is α_n -generic.

Proof. We use Proposition 1.11. Since α_{n+1} is much greater than α_n , the family $R_n \cup R_{n+1}$ is α_n -generic. Since R_A is hyperarithmetically generic, it is α_{n+1} -generic. Then the family $R_n \cup R_{n+1} \cup R_A$ is also α_n -generic. $\square$

We define forcing. We suppose that sets represented by elements of $C_{<n}$ have already been determined, and the set A has been determined for $k < n + 1$. We consider statements $\varphi(\bar{c}, a)$, with the meaning $\mathcal{N}_{n+1} \models \varphi(\bar{c}, a)$, for $\bar{c} \in C_{\leq n+1}$. We define forcing of these statements by our current forcing conditions. As before, the important clauses are the ones for basic statements.

Definition 2.4 (Definition of forcing).

1. For $k \leq n$, $p \Vdash U_k a$ if $k \in A$, and $p \Vdash \neg U_k a$ if $k \notin A$.
2. For $k > n$, $p \Vdash U_k a$ if $p(a, k) = 1$, and $p \Vdash \neg U_k a$ if $p(a, k) = 0$.
3. For $c \in C_{<n}$, $p \Vdash \pm U_k c$ if k is in, or not in, the set chosen for c to represent.
4. If $c \in C_n$ and $k < n$, or if $c \in C_{n+1}$ and $k < n + 1$, then $p \Vdash U_k c$ if $k \in A$, and $p \Vdash \neg U_k c$ if $k \notin A$.

5. If $c \in C_n$ and $k \geq n$, or if $c \in C_{n+1}$ and $k > n$, then $p \Vdash U_k c$ if $p(c, k) = 1$ and $p \Vdash \neg U_k c$ if $p(c, k) = 0$.

6. For more complicated formulas, forcing is defined as before.

Again, we have the important lemmas, including Truth-and-Forcing, Reduction to Basic Statements, and Transposition.

Lemma 2.23 (Truth and Forcing). *For a computable Σ_{α_n} formula $\varphi(\bar{u}, x)$ and $\bar{c} \in C_{\leq n+1}$, $\mathcal{N}_{n+1} \models \varphi(\bar{c}, a)$ iff there exists $p \subseteq \chi_{R_n \cup R_{n+1} \cup R_A}$ such that $p \Vdash \varphi(\bar{c}, a)$.*

Lemma 2.24 (Reduction to Basic Statements). *Suppose p and q are forcing conditions. If they force the same basic statements, then they force exactly the same computable Σ_{α_n} statements.*

Lemma 2.25 (Transposition). *Suppose f transposes $c \in C_{n+1}$ with $c' \in C_n$, fixing the rest of $C_{\leq n+1} \cup \{a\}$. Let p be a forcing condition such that the basic statements forced by p about the pair c, c' (and other constants), are forced by $f(p)$ about the pair c', c (and those other constants). Then for all computable Σ_{α_n} statements φ , $p \Vdash \varphi$ iff $f(p) \Vdash f(\varphi)$.*

Suppose, for example, that f transposes $c \in C_{n+1}$ with $c' \in C_n$, and p mentions c (along with a , and other constants $\bar{b}$), but not c' . Suppose that for $k \leq n$, $p(c, k)$ agrees with A . Then the basic statements forced by p about the pair c, c' match the ones forced by $f(p)$ about the pair c', c .

By the Tarski Criterion, to show that $\mathcal{N}_n \prec_{\alpha_n} \mathcal{N}_{n+1}$, it is enough to prove the following.

Proposition 2.6, re-phrased. Suppose $\mathcal{N}_{n+1} \models \varphi(\bar{d}, a, c)$, where φ is computable Σ_{α_n} , $\bar{d} \in C_{\leq n}$, and $c \in C_{n+1}$. Then there is some $c' \in C_n$ such that $\mathcal{N}_{n+1} \models \varphi(\bar{d}, a, c')$.

Proof. Suppose $\mathcal{N}_{n+1} \models \varphi(\bar{d}, c, a)$. By the Truth and Forcing Lemma, there is some $p \subseteq \chi_{R_n \cup R_{n+1} \cup R_A}$ such that $p \Vdash \varphi(\bar{d}, c, a)$. Let p' be the same as p except that for $k < n+1$, $p'(c, k)$ agrees with A . By the lemma on Reduction to Basic Statements, $p' \Vdash \varphi(\bar{d}, c, a)$. Let f be a permutation of $C_{\leq n+1} \cup \{a\}$ that transposes c with $c' \in C_n$, fixing everything else. We choose $c' \notin \bar{d}$, not mentioned in p or p' , such that $\chi_{R_n}(c', k) = p'(c, k)$ for all k for which $p'(c, k)$ is defined. By the Transposition Lemma, we have $f(p') \Vdash \varphi(\bar{d}, c', a)$. Moreover, $f(p') \subseteq \chi_{R_n \cup R_A}$. By the Truth and Forcing Lemma, $\mathcal{N}_{n+1} \models \varphi(\bar{d}, a')$, as required. $\square$

We have finished the proof of the three propositions that were needed for Proposition 2.3. This completes the proof of Theorem 2.1.

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